Takaya Nakashima, M.D.

Anesthesiologist and biostatistician

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Research profile

I study treatment timing and patient trajectories in sepsis and perioperative care. My work uses clinical databases and multicenter registries, including MIMIC-IV, eICU-CRD, JIPAD and J-CKD-DB-Ex. I work on study design and statistical analysis at the Clinical Research Center, Nagasaki University Hospital.

Appointments

April 2026 - present | Research Associate
Clinical Research Center, Nagasaki University Hospital

April 2024 - present | Physician
Department of Anesthesiology and Intensive Care Medicine, Nagasaki University Hospital

April 2022 - March 2024 | Junior Resident
Nagasaki University Hospital

Current | Collaborative Researcher
TXP Medical Co. Ltd.

Education

April 2022 - present | Doctoral program in Anesthesiology and Intensive Care Medicine
Graduate School of Biomedical Sciences, Nagasaki University. Degree in progress.

April 2016 - March 2022 | M.D., School of Medicine
Nagasaki University

Methods and software

Target trial emulation; clone-censor-weight estimation; marginal structural models; survival analysis; nonlinear mixed-effects models; functional data analysis; clinical prediction and machine learning. Software: R and Python.

Japan Statistical Society Certificate, Grade Pre-1

Professional memberships

April 2024 - present | Japanese Society of Anesthesiologists
September 2022 - present | Japan Epidemiological Association
September 2021 - present | Japanese Society for Clinical Epidemiology
June 2021 - present | Japanese Association for Medical AI
November 2020 - present | Behaviormetric Society of Japan

Journal publications

Author order follows the publication record. First and co-first authorship are marked where applicable. Corresponding authorship is marked for the two 2026 critical care papers.

1. **Nakashima T**, Sekino M, Araki H, Ichinomiya T, Hara T. Clinical characteristics, management, and outcomes of adults admitted to participating Japanese intensive care units with registry-identified anaphylactic shock: a multicenter retrospective descriptive study. *Journal of Anesthesia, Analgesia and Critical Care*. Published online October 30, 2026.
doi: 10.1186/s44158-026-00464-x | First and corresponding author

2. **Nakashima T**, Ichinomiya T, Nakajima M, Shinozaki T, Shibata J, Goto T, Sato S, Hara T. Ultra-early versus early adjunctive vasopressin initiation after norepinephrine escalation in septic shock: a target trial emulation. *Intensive Care Medicine*. 2026;52(10):2085-2095.
doi: 10.1007/s00134-026-08575-3 | Analysis code | First and corresponding author

3. **Nakashima T**, Nagasu H, Hirano A, Iwagami M, Goto T, Kashihara N, Nangaku M, Hirakawa Y. The eGFR difference between creatinine-based and cystatin-C-based equations is also important in an Asian population within the J-CKD-DB-Ex. *Scientific Reports*. 2026;16:26286.
doi: 10.1038/s41598-026-57444-y | First author

4. Nagasu H, **Nakashima T**, Ihara K, Fujimori R, Goto T, Nitta D, Kishi S, Sasaki T, Kashihara N. Prediction of estimated glomerular filtration rate slope and kidney prognosis of patients with chronic kidney disease. *Scientific Reports*. 2026;16:8883.
doi: 10.1038/s41598-026-38246-8 | Co-first author

5. Kashihara N, Itano S, **Nakashima T**, Goto T, Yoshihara K, Eguchi S, Iekushi K, Isaka Y, Nagasu H; J-CKD-DB collaborators. Proteinuria reduction as a surrogate endpoint for clinical study of IgA nephropathy in Japanese patients: data from the J-CKD-DB-Ex. *Clinical and Experimental Nephrology*. 2026;30(2):309-319. Published online 2025.
doi: 10.1007/s10157-025-02788-4 | Co-first author

6. Nakahashi S, Suzuki K, **Nakashima T**, Hayashi Y, Tanabe Y, Tanaka A, Hashiuchi S, Yamashita C, Ito Y, Wada T, et al. A reappraisal of association between ventilator-associated events and mortality among critically ill patients using marginal structural model: multicenter observational study. *Intensive Care Medicine*. 2025;51(10):1764-1774.
doi: 10.1007/s00134-025-08074-x

7. Liu K, **Nakashima T**, Goto T, Nakamura K, Nakano H, Motoki M, Kamijo H, Matsuoka A, Ishii K, Morita Y, et al. Phenotypes of Functional Decline or Recovery in Sepsis ICU Survivors: Insights From a 1-Year Follow-Up Multicenter Cohort Analysis. *Critical Care Medicine*. 2025;53(4):e928-e940.
doi: 10.1097/CCM.0000000000006621 | Co-first author

Journal publications (continued)

8. **Nakashima T**, Sato S, Matsui H, Mizuno A. Estimating the Timeframe of Lactate Reduction in Acute Cardiovascular Disease Using Functional Data Analysis Based on Nonlinear Mixed Effects Models. *International Heart Journal*. 2024;65(6):1058-1065.
doi: 10.1536/ihj.23-659 | First author

9. Kishi S, **Nakashima T**, Goto T, Nagasu H, Brooks CR, Okada H, Tamura K, Nakano T, Narita I, Maruyama S, et al. Association of serum magnesium levels with renal prognosis in patients with chronic kidney disease. *Clinical and Experimental Nephrology*. 2024;28(8):784-792.
doi: 10.1007/s10157-024-02486-7 | Co-first author

10. Kasugai D, **Nakashima T**, Goto T. Clinical implications of urine output-based sepsis-associated acute kidney injury. *Intensive Care Medicine*. 2023;49(10):1263-1265.
doi: 10.1007/s00134-023-07190-w

11. **Nakashima T**, Onagase Y, Fukahori S, Fukushima C, Iriki H, Nagae Y, Yamaguchi H, Sakamoto N, Ishimatsu Y, Yamada R, et al. Potential of a lung sound analysis system using a new recording device and signal separation as an aid to physician auscultation. *Nagasaki Medical Journal*. 2022;97(1):1-6. In Japanese.
Publication record | First author

12. Kaneko S, Hara K, Sato S, **Nakashima T**, Kawazoe Y, Taguchi M, Urabe S, Nakao A, Hamada K, Yamaguchi M, Hara T. Association between preoperative toe perfusion index and maternal core temperature decrease during cesarean delivery under spinal anesthesia: a prospective cohort study. *BMC Anesthesiology*. 2021;21:250.
doi: 10.1186/s12871-021-01470-y

Conference abstract

Nagasu H, **Nakashima T**, Ihara K, Fujimori R, Goto T, Nitta D, Kishi S, Sasaki T, Kashihara N. Development of an eGFR slope prediction model using J-CKD-DB-Ex and visualization of risk with a web application. Japanese Journal of Nephrology. 2025;67(4):484. In Japanese. Abstract record

Research tools

eGFR slope prediction web application
Prediction and visualization of eGFR slope and kidney outcomes using J-CKD-DB-Ex data. Hosted by the Japan Kidney Association. Prediction application

Lung sound separation and analysis
Recorded lung sounds were separated and analyzed to examine changes in coarse crackles before and after dialysis. Related publication listed above. Publication record

Books, chapters, translations and articles

Japanese-language outputs; English titles below are translations for this CV.

2026.06 | Generative AI in medical writing: a practical guide to accelerating study planning and literature review
Respiratory Therapy (Jpn Soc Respiratory Care Med) / Sole author

2025.06 | Residents Note Vol.27 No.6: Gathering and using medical information
Yodosha / Chapter: Reading papers in the generative AI era

2025.03.19 | TRIPOD+AI Expanded Checklist: volunteer Japanese translation, version 1.1
Co-translators: Shuntaro Sato, Takaya Nakashima, Ryunosuke Goto, and Tadahiro Goto. Hosted on the TRIPOD website. Date of the Japanese version.

2024.12 | Report on the 29th Annual Meeting of the Japanese Society of Cardiovascular Anesthesiologists
Circulation Control 45(2): 130-131. Takaya Nakashima. Japan Society of Circulation Control in Medicine.

2024.09-2025.01 | Igakukai Shimbun Plus (series): Let the machines do the tedious work
Igaku-Shoin, Igakukai Shimbun Plus. Ten-part series.

2024.05 | Modern Epidemiology, 4th ed. (Japanese edition)
Gakujutsu Tosho / Co-translator, Ch.18 Stratification and Standardization

2023.09 | AI-assisted literature search and management for busy residents
Shukan Igakukai Shimbun (Resident issue) No.3532

Teaching

2026 - present | Medical Statistics (School of Dentistry)
Nagasaki University

On-demand course | AI-assisted research and paper writing for busy clinicians
Instructor, mJOHNSNOW (mMEDICI). Five sessions. Generative AI and digital tools for exploring research topics, searching and managing literature, statistical analysis, paper writing, and slide preparation.

Awards

2026.09 | Best Presentation Award, 64th Annual Meeting of the Kyushu Society of Anesthesiologists
Japanese Society of Anesthesiologists, Kyushu Branch

2021.10 | Young Investigator Award (Clinical Research)
87th Kyushu Branch Autumn Meeting, Japanese Respiratory Society and affiliated societies

Invited lectures and conference presentations

Titles translated from Japanese. The SAS entry records the published program listing.

2026.10.13 | Benefits and challenges of generative AI assistance in statistical analysis practice
SAS User Meeting 2026, Day 0 session on statistical analysis in academia. With Shuntaro Sato. Listed in the conference program.

2026.10.07 | Clinical researchers in the AI era: how will the value of researchers change?
Invited lecture, Kochi Medical Revitalization Organization rheumatology specialist training program, Kochi University.

2026.09.19 | Clinical researchers in the AI era: working with generative AI
Educational Lecture 2, 31st Annual Meeting, Japanese Society of Cardiovascular Anesthesiologists. On-demand, September 19 - November 25, 2026.

2026.09.05 | Ultra-early versus early adjunctive vasopressin initiation after norepinephrine escalation in septic shock: a retrospective observational target trial emulation using two ICU databases
64th Annual Meeting, Kyushu Society of Anesthesiologists. Best Presentation Award.

2025.07.24 | Using AI tools to write papers efficiently: tips for busy researchers
Clinical Research Seminar, Clinical Research Center, Nagasaki University Hospital. Online.

2024.03 | Causative organism and outcome in bacteremia requiring intensive care: a retrospective cohort study
51st Annual Meeting, Japanese Society of Intensive Care Medicine

2023.09 | Predicting hemodynamic stabilization from lactate trajectories: functional data analysis with mixed effects models
Japanese Federation of Statistical Science Associations 2023

2023.02 | A prediction model for time to cardiovascular events in cardiac sarcoidosis
33rd Annual Meeting, Japan Epidemiological Association

2022.11 | Predicting recovery time in acute cardiovascular disease using functional data based on nonlinear mixed effects models
5th Annual Meeting, Japanese Society for Clinical Epidemiology

2021.10.22 | Clinical application of separation-technique analysis of recorded lung sounds to changes in coarse crackles before and after dialysis
87th Kyushu Branch Autumn Meeting, Japanese Respiratory Society and affiliated societies. Abstract 074, Kumamoto Castle Hall.